IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Satish Kumar Ram
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Objective**
  - Predict Falcon 9 first-stage landing success using historical launch data and machine learning.
- **Summary of Methodology**
  - Data Collection through **SpaceX REST API.**
  - Data Collection using **Wikipedia Web Scraping.**
  - Data Wrangling and Feature Engineering.
  - Exploratory Data Analysis (EDA) using SQL Queries, Python Visualizations (Matplotlib & Seaborn), and Plotly Dash.
  - Interactive Visual Analytics using **Folium** and **Plotly Dash.**
  - Predictive Analysis using **Classification Models.**
- **Summary of Results**
  - Landing success rate improved significantly over time.
  - Payload mass, orbit type, and launch site influenced landing success.
  - SQL and visualizations revealed important launch patterns.
  - Interactive dashboards provided deeper insights into launch data.
  - The best classification model achieved the highest prediction accuracy.

Introduction

- **Project Background and Context:**
  - SpaceX has significantly reduced the cost of space missions by successfully recovering and reusing Falcon 9 first-stage boosters. Traditional rocket launches are expensive because rockets are usually discarded after launch. By predicting whether the Falcon 9 first stage will land successfully, organizations can estimate mission costs, improve planning, and support data-driven decision-making. This project uses historical SpaceX launch data to analyze landing outcomes and build machine learning models for landing prediction.
- **Project Objective:**
  - The project aims to identify the factors influencing Falcon 9 first-stage landing success and develop a machine learning model to predict landing outcomes.
  - Collect Falcon 9 launch data using the SpaceX REST API and **Wikipedia Web Scraping**.
  - Perform **data wrangling, SQL analysis, and exploratory data analysis (EDA).**
  - Build interactive visualizations using SQL, Folium, and Plotly Dash.
  - Develop, evaluate, and compare machine learning classification models to predict Falcon 9 first-stage landing success.

Section 1

Methodology

Methodology

- **Data Collection Methodology**
  - Collected Falcon 9 launch data from the SpaceX REST API.
  - Extracted additional launch records using Wikipedia web scraping.
- **Data Wrangling**
  - Cleaned missing values and standardized the dataset.
  - Created the landing success label and prepared features for analysis.
- **Exploratory Data Analysis (EDA)**
  - Performed SQL queries to analyze launch statistics.
  - Created visualizations using Matplotlib and Seaborn.
- **Interactive Visual Analytics**
  - Built an interactive Folium map to analyze launch site locations.
  - Developed a Plotly Dash dashboard with filters and interactive charts.
- **Predictive Analysis**
  - Built and evaluated Logistic Regression, SVM, Decision Tree, and KNN models.
  - Selected the Decision Tree classifier as the best-performing model based on accuracy.

Data Collection

- Falcon 9 launch data was collected using the SpaceX REST API and Wikipedia web scraping.
- Retrieved Falcon 9 launch data using HTTP GET requests to the SpaceX REST API.
- Converted the JSON response into a Pandas DataFrame using .json() and pd.json_normalize().
- Cleaned the dataset by checking and handling missing values where required.
- Collected additional Falcon 9 launch records from Wikipedia using BeautifulSoup.
- Extracted HTML tables, parsed the launch records, and converted them into a Pandas DataFrame for further analysis.

Data Collection – SpaceX API

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Retrieved launch details including launch date, payload mass, orbit, launch site, booster version, and landing outcome.
- Converted the JSON response into a structured Pandas DataFrame using pd.json_normalize().
- Verified successful API response (HTTP 200) and prepared the dataset for analysis.

```python
import requests
import pandas as pd

static_json_url = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json"

response = requests.get(static_json_url)

print(response.status_code)
```

```
200
```

We should see that the request was successfull with the 200 status response code

```python
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```python
# Use json_normalize method to convert the json result into a dataframe
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```python
# Get the head of the dataframe
data.head(5)
```

Export to csv

Data Collection - Scraping

- Extracted additional Falcon 9 launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted the extracted records into a structured **Pandas DataFrame**.
- Used the scraped data to enrich and validate the launch dataset.
- Prepared the scraped launch records for further data wrangling and analysis.

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(static_url, headers=headers)
```

Create a `BeautifulSoup` object from the HTML `response`

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content

soup = BeautifulSoup(response.text, "html.parser")
```

```
print(soup.title.string)
```

```
List of Falcon 9 and Falcon Heavy launches - Wikipedia
```

Print the page title to verify if the `BeautifulSoup` object was created properly

```
# Use soup.title attribute
print(soup.title)
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`

html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

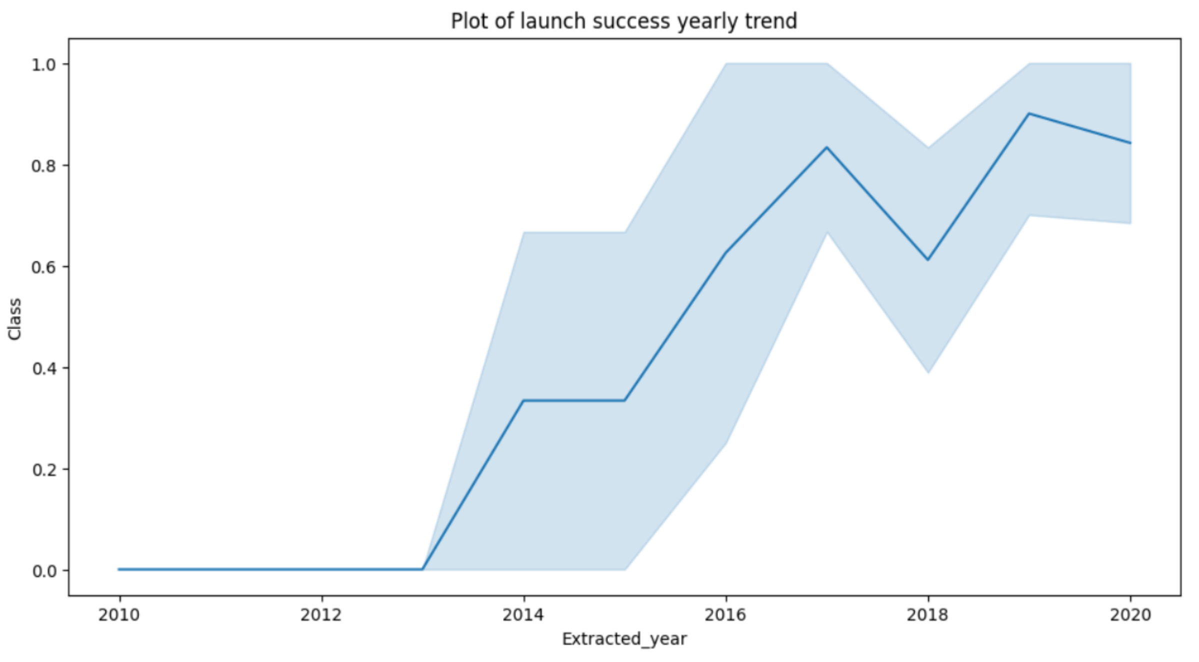
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
print(first_launch_table)
```

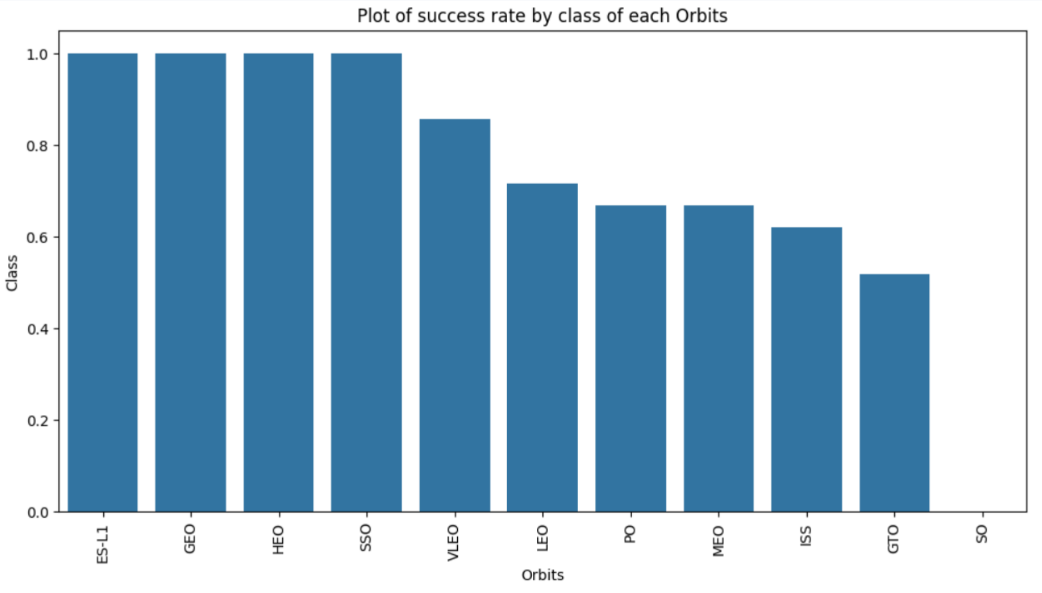
Data Wrangling

- Removed missing and inconsistent values from the dataset.
- Standardized column names and data formats for consistency.
- Created the **landing_class** target variable from the mission outcome.
- Selected relevant features for exploratory analysis and machine learning.
- Exported the cleaned dataset for EDA and predictive modelling.

EDA with Data Visualization

- Launch success rate showed a steady improvement over the years.
- Success rates varied across different orbit types, indicating mission characteristics affect landing outcomes.
- Visual analysis helped identify trends and relationships for predictive modelling.





GitHub Repository:

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EDA with SQL

- Used **SQL queries** to analyze SpaceX Falcon 9 launch data.
- Retrieved launch statistics by **launch site, mission outcome, booster version, and payload mass**.
- Identified **successful landing patterns** and launch frequency across different sites.
- Generated insights that supported the **EDA and predictive analysis**.

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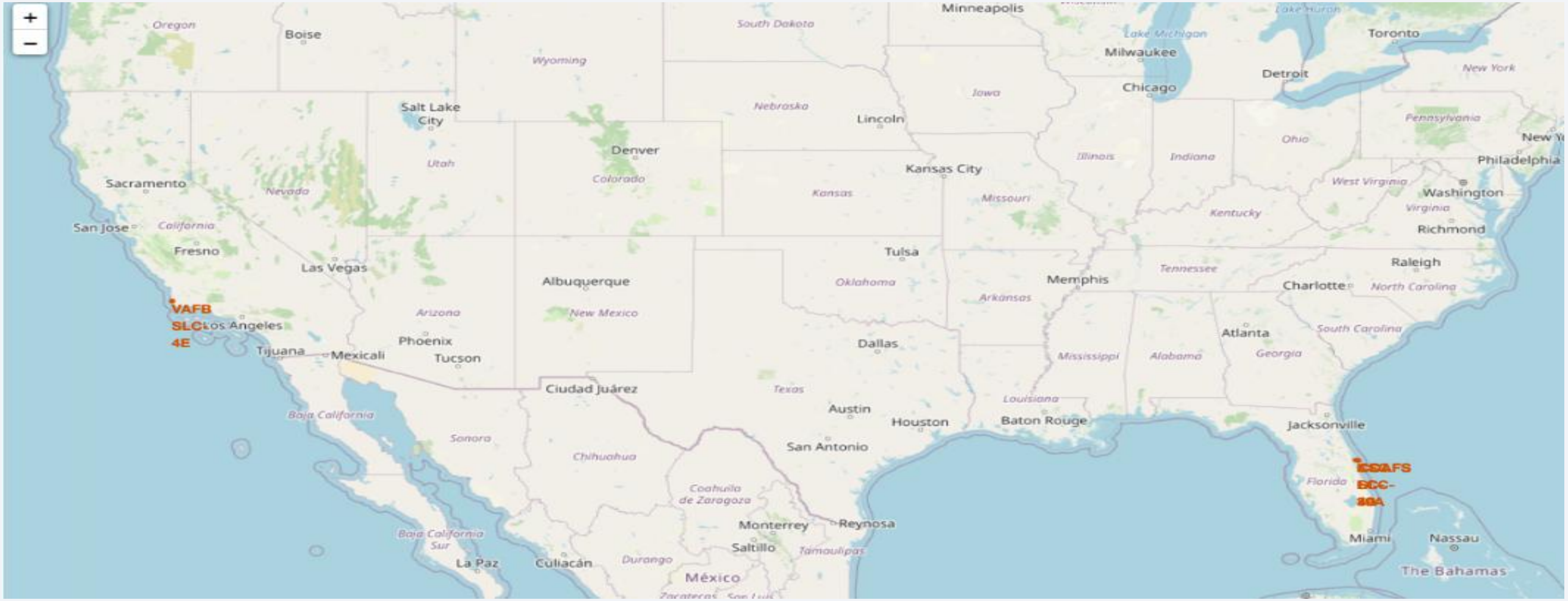
Interactive Map with Folium

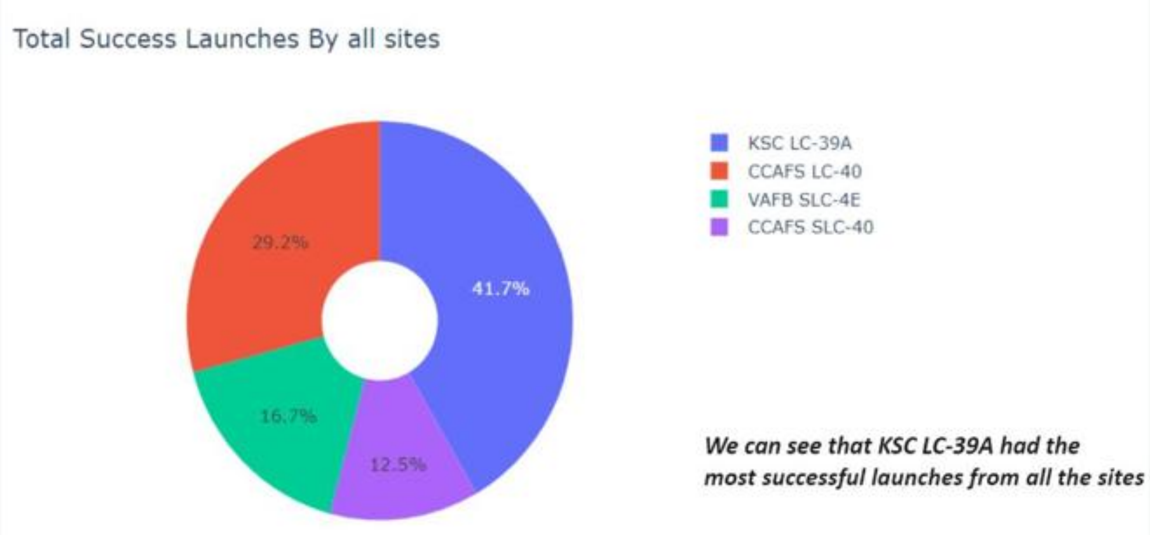
- Created an interactive **Folium map** to visualize SpaceX launch site locations.
- Displayed launch sites using **markers and geographic coordinates**.
- Explored the relationship between **launch site locations and mission outcomes**.
- The Folium map enabled interactive exploration of launch site locations through zoom and pan features.



Build a Dashboard with Plotly Dash

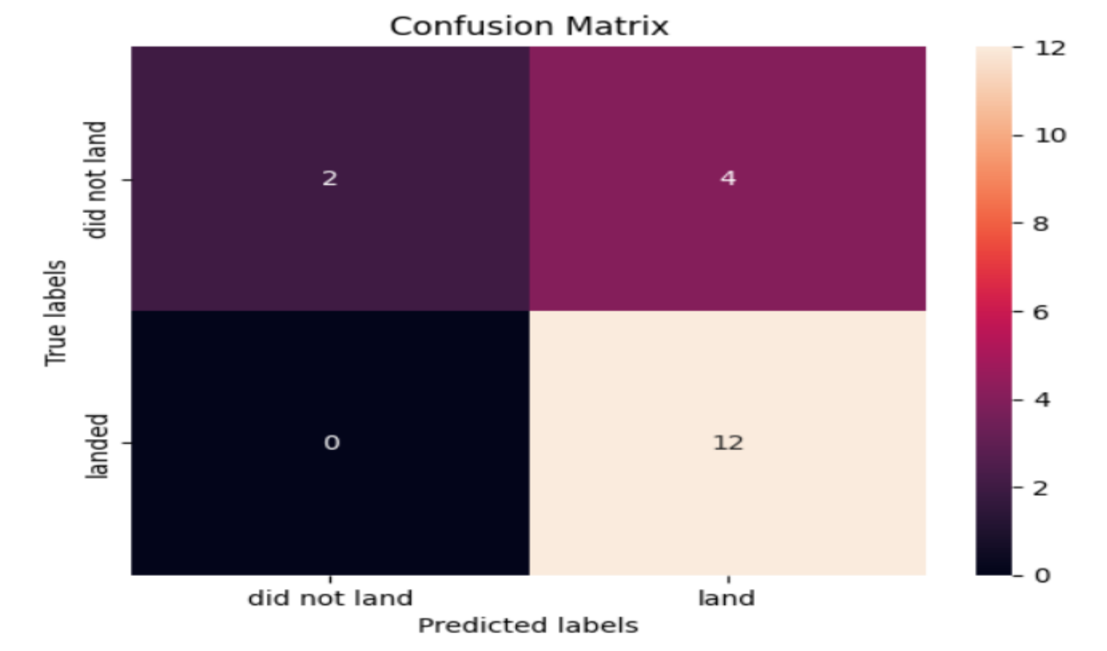
- Developed an interactive Plotly Dash dashboard to analyze Falcon 9 launch success across different launch sites and payload ranges.
- Implemented **launch site filters** using dropdown menus for dynamic data exploration.
- Created **pie charts** to visualize launch success across different launch sites.
- Built a scatter plot to analyze the relationship between payload mass and landing success, with booster version categories represented by different colors.

Figure: Plotly Dash dashboard output showing launch success distribution by launch site.



Predictive Analysis (Classification)

Logistic Regression Confusion Matrix




```
print("Logistic Regression:", logreg_cv.score(X_test, Y_test))
print("SVM:", svm_cv.score(X_test, Y_test))
print("Decision Tree:", tree_cv.score(X_test, Y_test))
print("KNN:", knn_cv.score(X_test, Y_test))
```

```
Logistic Regression: 0.8333333333333334
SVM: 0.8333333333333334
Decision Tree: 0.9444444444444444
KNN: 0.7777777777777778
```

- Built and evaluated **Logistic Regression, SVM, Decision Tree, and KNN** models.
- Compared model performance using **test accuracy and confusion matrix**.
- **Decision Tree achieved the highest prediction accuracy (94.44%)** among all models.
- Selected Decision Tree as the best-performing predictive model.

Evaluate Each Model

Results

- **Project Summary:** Integrated data collection, EDA, SQL, interactive dashboards, and machine learning to predict Falcon 9 landing success.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis: Decision Tree achieved the highest prediction accuracy of 94.44%.**
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 landing success and support mission planning.

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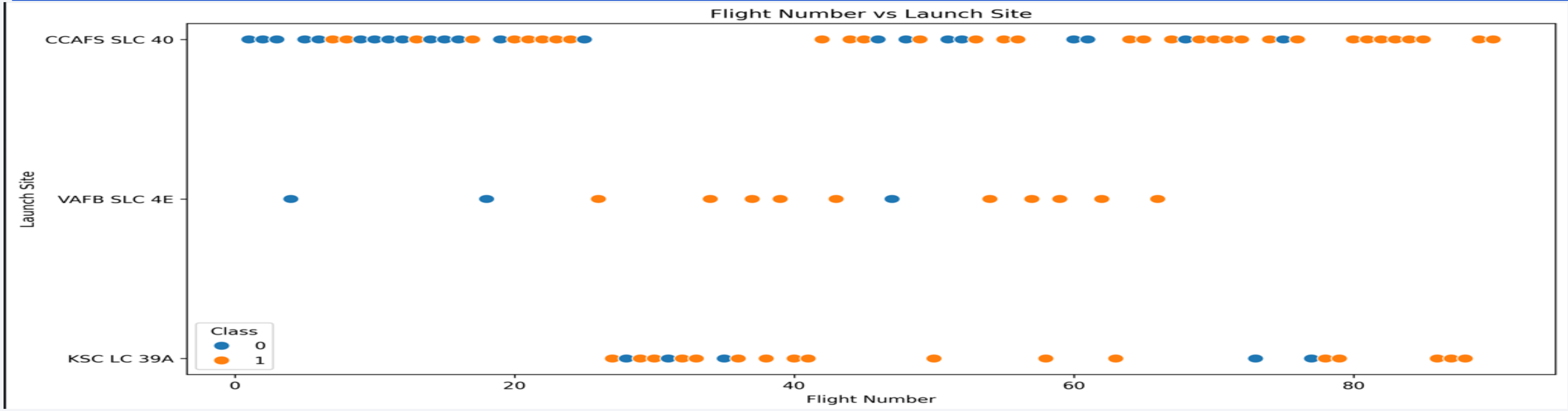
Section 2

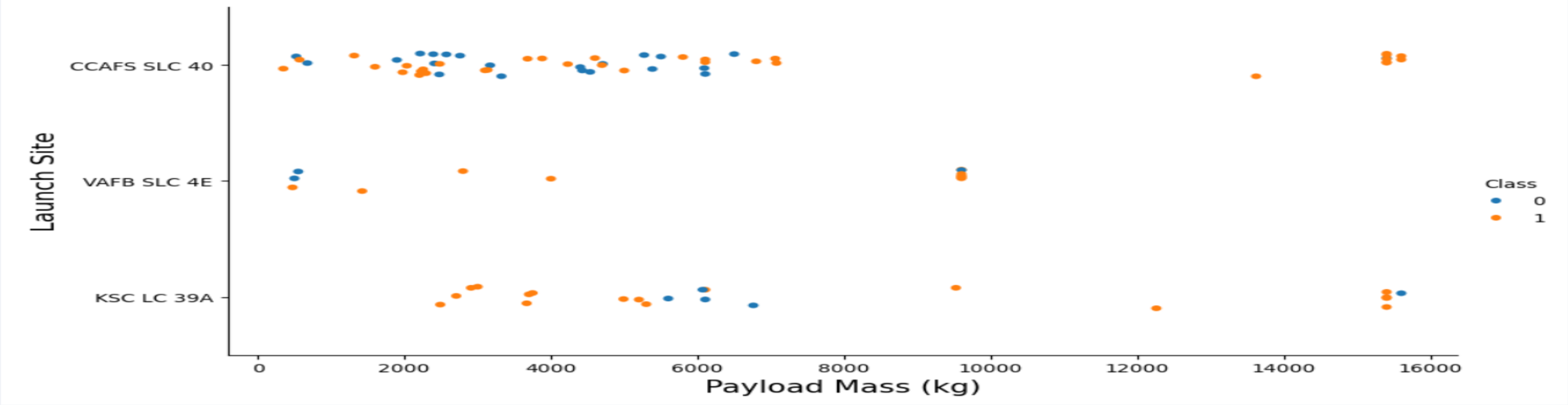
Insights drawn from EDA

Flight Number vs. Launch Site



- Early Falcon 9 missions had lower landing success, while later missions showed higher success rates.
- Launch success improved as the **flight number** increased, indicating continuous improvements in booster recovery and operational experience.
- **CCAFS SLC 40** recorded the highest number of Falcon 9 launches, making it the most frequently used launch site.
- **KSC LC 39A** and **VAFB SLC 4E** also demonstrated successful landings, although with fewer launches than CCAFS SLC 40.
- Overall, the plot suggests that both operational experience (represented by flight number) and launch site are associated with Falcon 9 first-stage landing success.

Payload vs. Launch Site



- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, landing success varied across launch sites and payload masses, indicating that multiple factors contributed to Falcon 9 first-stage landing outcomes.

Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO achieved a 100% landing success rate in the available dataset.
- GTO missions recorded the lowest landing success rate (approximately 52%).
- LEO, ISS, MEO, PO, and VLEO showed intermediate success rates, indicating varying landing performance across orbit types.
- Overall, the results suggest an association between orbit type and Falcon 9 first-stage landing success.

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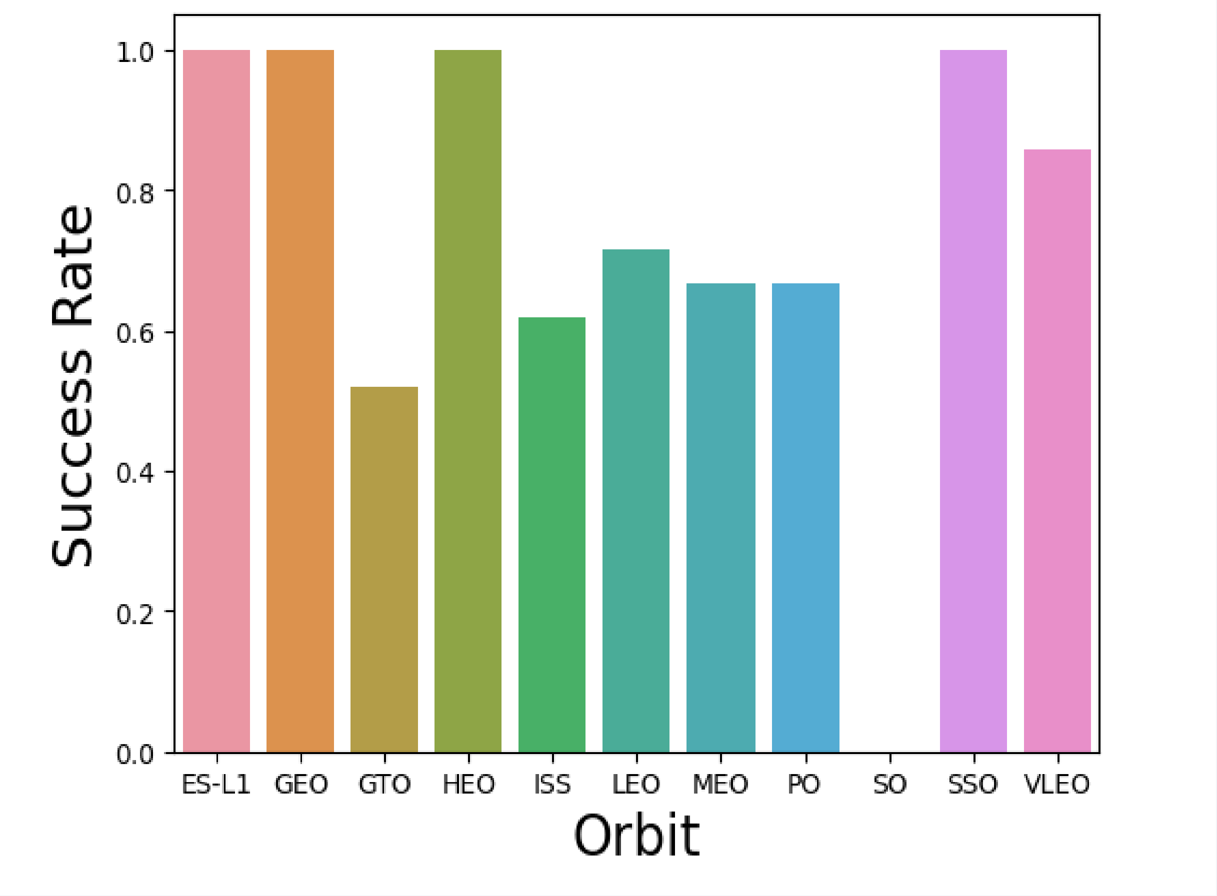
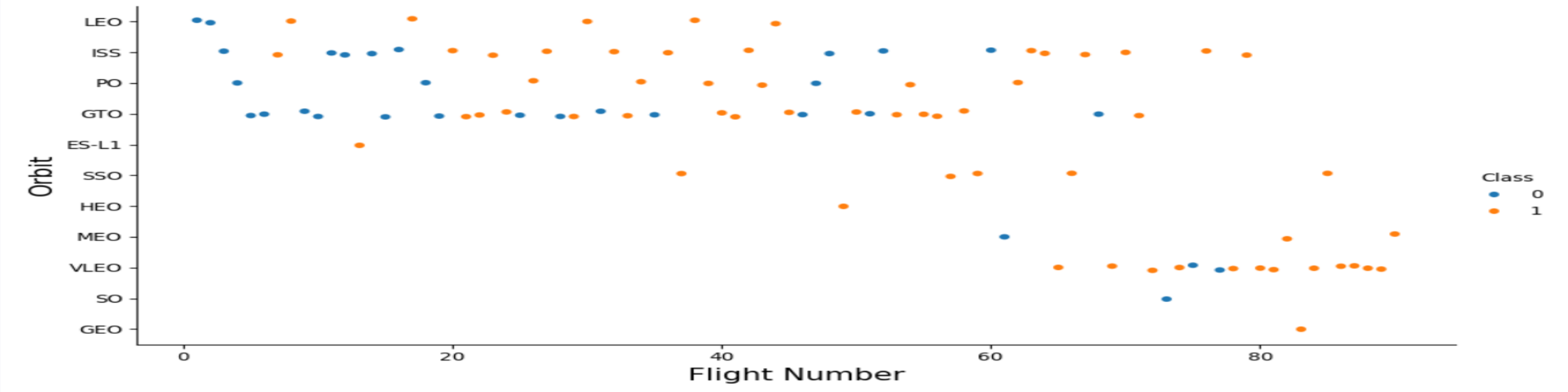
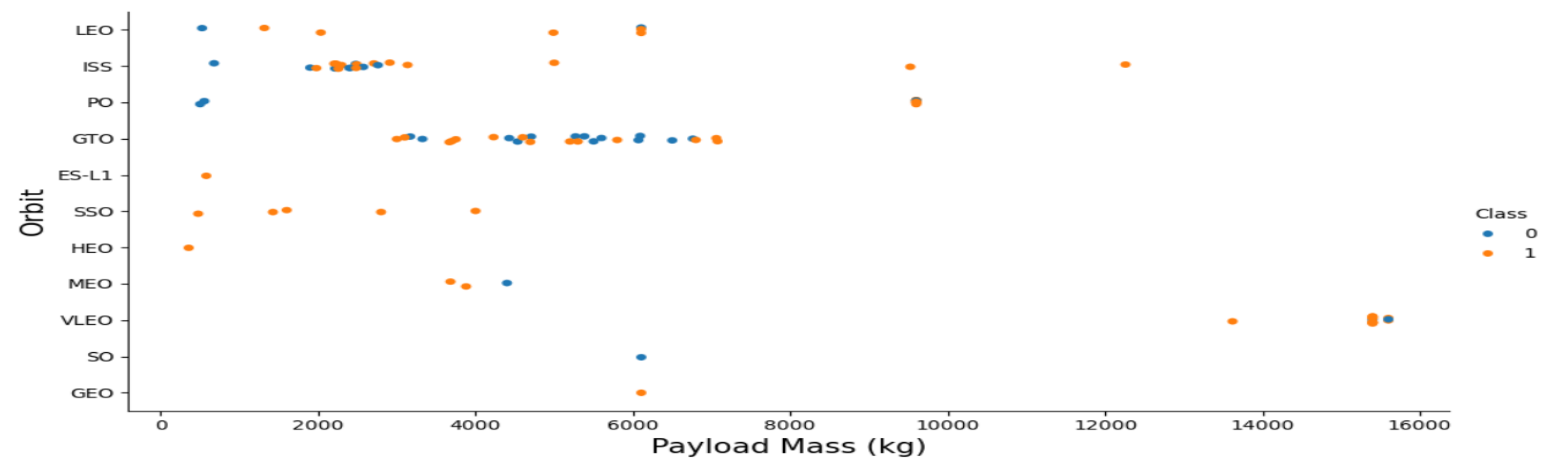


Flight Number vs. Orbit Type



- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Overall, both flight number and orbit type were associated with Falcon 9 first-stage landing success.

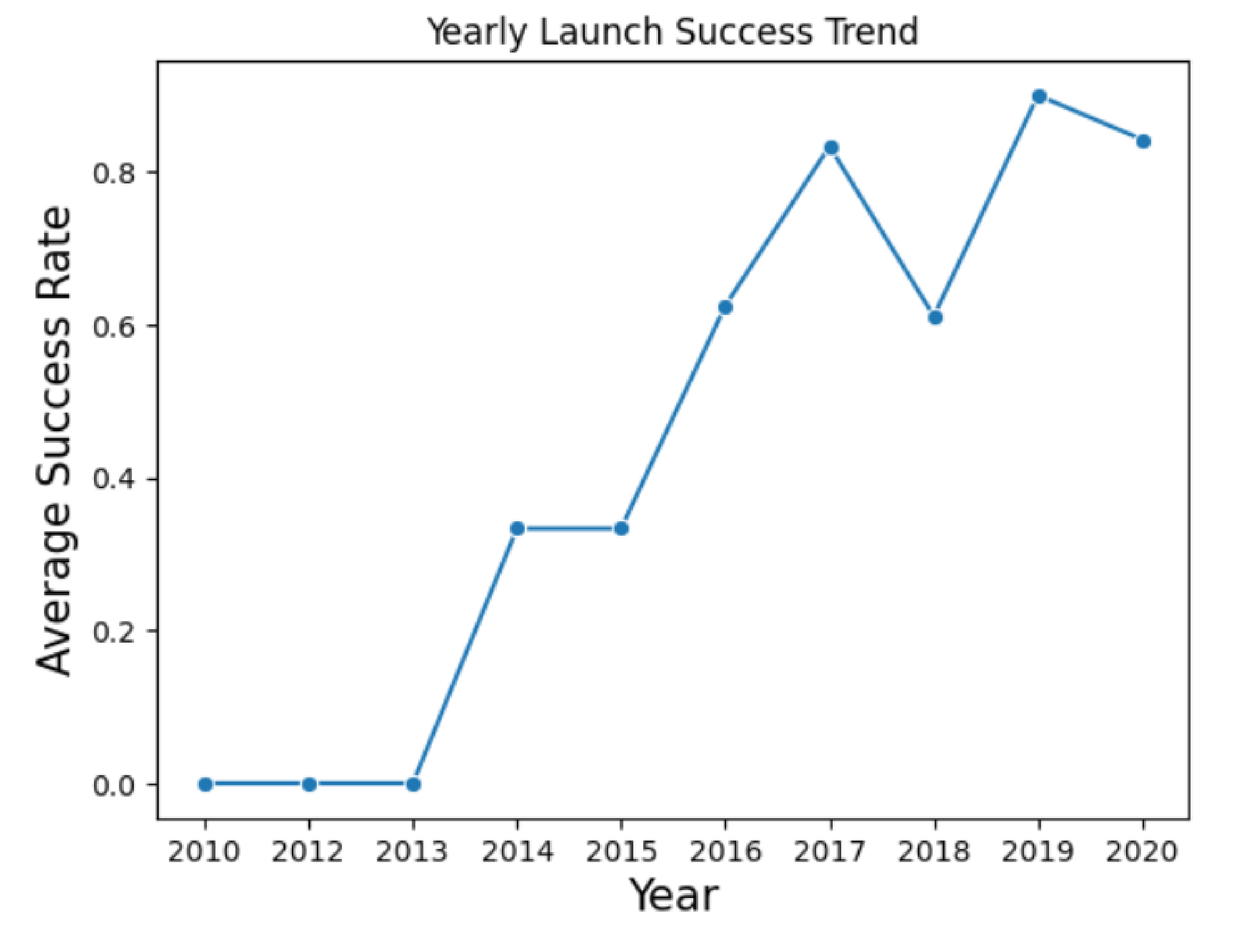
Payload vs. Orbit Type



- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low- to medium-payload missions.
- Overall, both payload mass and orbit type were associated with landing outcomes, although payload mass alone did not determine landing success.

Launch Success Yearly Trend

- Overall, first-stage landing success rates improved over the years despite minor fluctuations.
- Landing success rates were minimal during the early years of Falcon 9 operations (2010–2013).
- From 2016 onward, average landing success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high first-stage landing success rates, demonstrating improved booster recovery and operational reliability.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
%%sql

SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- **Identified four unique SpaceX launch sites from the dataset:**
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E
- Most launches were conducted from Cape Canaveral and Kennedy Space Center.
- Vandenberg Space Force Base (SLC-4E) primarily supported polar and sun-synchronous missions.
- These launch sites were used for EDA, dashboard development, and predictive modeling.

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql

SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The query returned launch records from Cape Canaveral launch sites (CCAFS).
- The first five matching records correspond to early Falcon 9 missions.
- SQL filtering using the LIKE operator helped identify launch-site-specific records efficiently.

Total Payload Mass

- The SQL query calculated a total payload mass of 45,596 kg for NASA (CRS) missions.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql

SELECT SUM(Payload_Mass__kg_) AS Total_Payload_Mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Total_Payload_Mass
45596

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

[22]:
```
%%sql

SELECT Booster_Version,
       AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
GROUP BY Booster_Version;
```

```
 * sqlite:///my_data1.db
Done.
```

[22]:

Booster_Version	Average_Payload_Mass
F9 v1.1	2928.4

- The Falcon 9 v1.1 booster carried an average payload mass of **2928.4 kg**, which serves as a reference for comparing later booster versions.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

[24]:
```
%%sql

SELECT MIN(Date) AS First_Successful_Landing_Date
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

[24]:

First_Successful_Landing_Date
2015-12-22

- The first successful ground pad landing occurred on **December 22, 2015**.
- This event demonstrated the successful recovery of a Falcon 9 first-stage booster on land.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[26]:
```
%%sql

SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND Payload_Mass__kg_ > 4000
  AND Payload_Mass__kg_ < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

[26]:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Four Falcon 9 booster versions successfully landed on a drone ship while carrying payloads between 4,000 kg and 6,000 kg. This demonstrates SpaceX's successful recovery capability for medium-payload missions.

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

[55]:
```
import pandas as pd

task_7a = '''
SELECT COUNT(Mission_Outcome) AS SuccessOutcome
FROM SPACEXTABLE
WHERE Mission_Outcome LIKE 'Success%';
'''

pd.read_sql_query(task_7a, conn)
```

[55]:

	SuccessOutcome
0	100

[56]:
```
%%sql

SELECT AVG(Payload_Mass__kg_)
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

[56]:

AVG(Payload_Mass__kg_)
6138.287128712871

[58]:
```
task_7b = '''
SELECT COUNT(Mission_Outcome) AS FailureOutcome
FROM SPACEXTABLE
WHERE Mission_Outcome LIKE 'Failure%';
'''

pd.read_sql_query(task_7b, conn)
```

[58]:

	FailureOutcome
0	1

- **Key Findings:**
- A total of **100 successful mission outcomes** were recorded in the dataset.
- Only **1 mission outcome** was classified as a **failure**, indicating a very low mission failure rate.
- The results demonstrate the **high reliability and operational success** of SpaceX launch missions.

Boosters Carrying Maximum Payload

```
%%sql

SELECT Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- **Multiple Falcon 9 Block 5 booster versions** carried the **maximum payload mass of 15,600 kg**.
- The results indicate that **Falcon 9 Block 5** consistently supported SpaceX's **highest payload capacity**.
- These boosters demonstrate the capability of handling **heavy payload missions** while maintaining operational reliability.

2015 Launch Records

```
%%sql

SELECT
    substr(Date, 6, 2) AS Month,
    substr(Date, 1, 4) AS Year,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND Date LIKE '2015-%';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Year	Landing_Outcome	Booster_Version	Launch_Site
01	2015	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	2015	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- **Two drone ship landing failures** were recorded during **2015**.
- Both failed landings occurred at **CCAFS LC-40** using **Falcon 9 v1.1** booster versions.
- These missions provided valuable operational insights that contributed to improvements in subsequent landing attempts.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **"No attempt"** was the most frequent landing outcome with **10 occurrences** during the selected period.
- **Success (ground pad)**, **Success (drone ship)**, and **Failure (drone ship)** each occurred **5 times**.
- Less frequent outcomes included **Controlled (ocean)**, **Uncontrolled (ocean)**, **Precluded (drone ship)**, and **Failure (parachute)**, showing the evolution of SpaceX landing strategies over time.

```
%%sql

SELECT Landing_Outcome,
       COUNT(Landing_Outcome) AS Total_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Total_Count DESC;
```

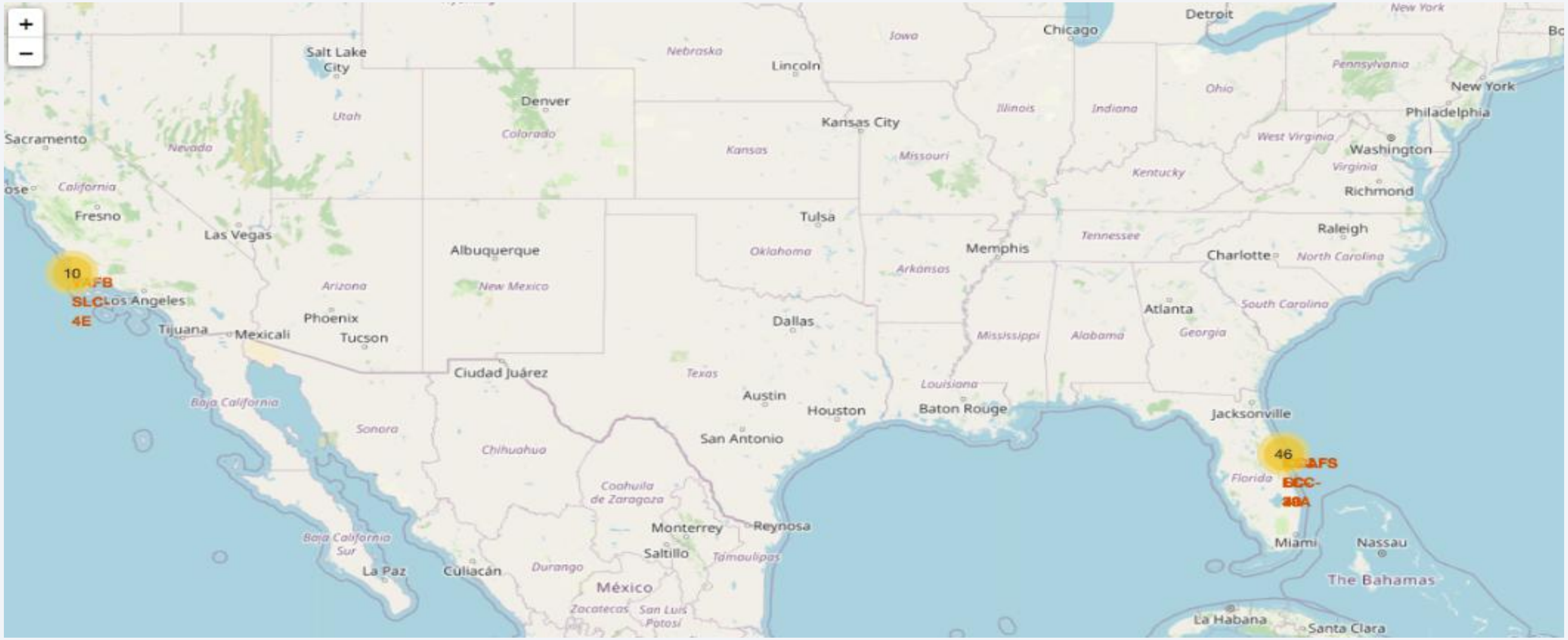
```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

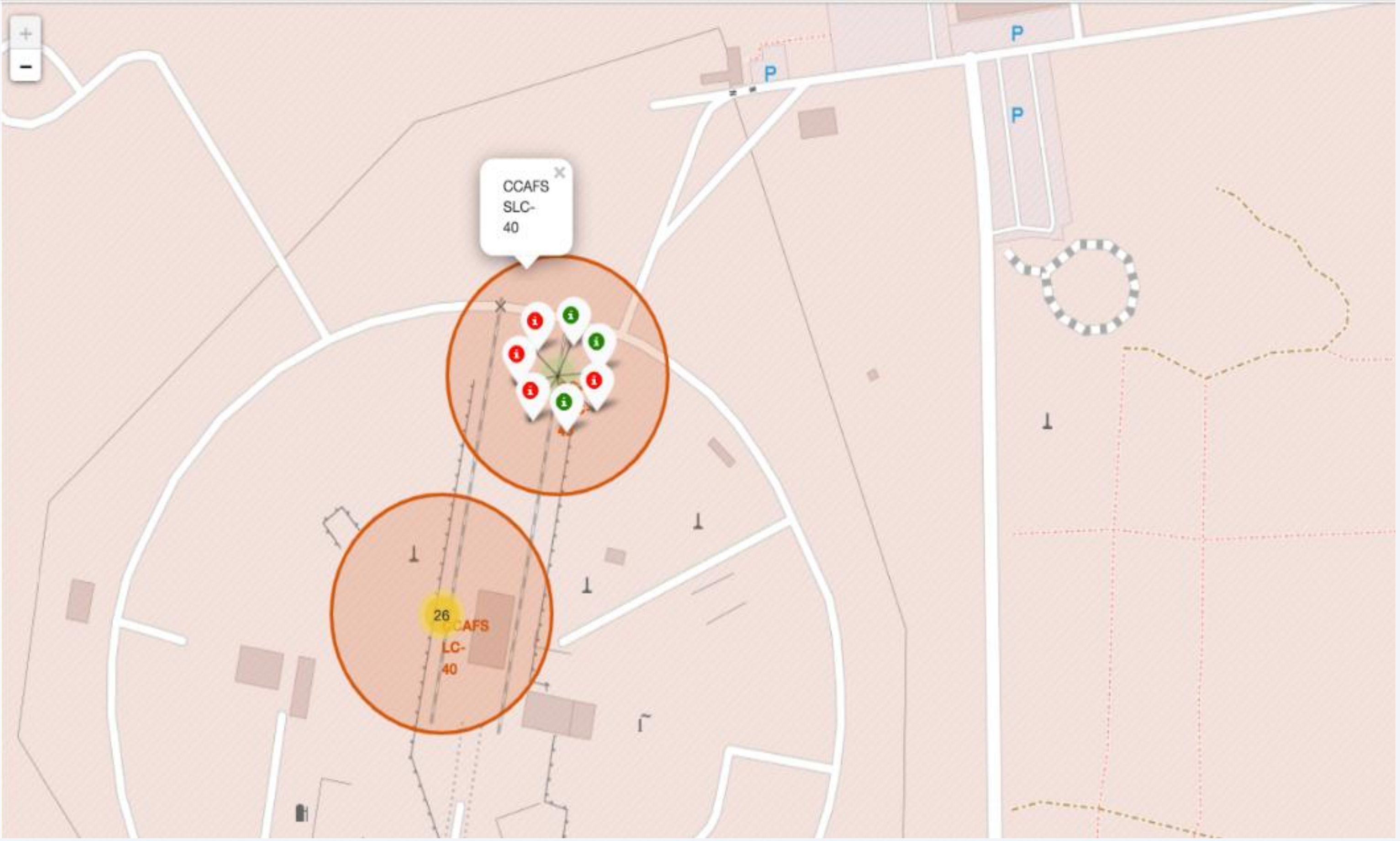
Launch Sites Proximities Analysis

Launch Site Locations



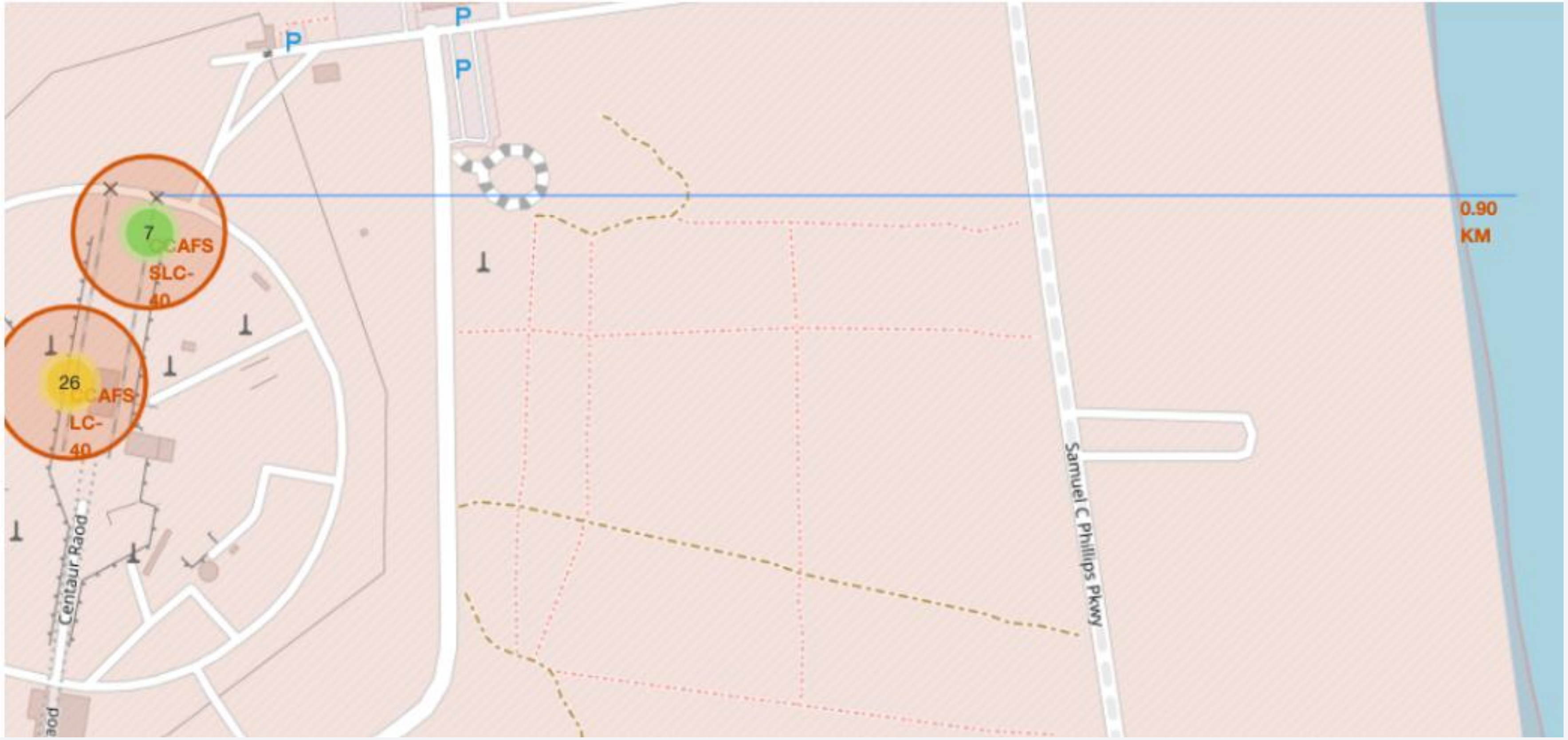
- All launch sites are located in **coastal areas** for safe rocket launches.
- **Vandenberg (California)** is the only West Coast launch site.
- **CCAFS** and **KSC (Florida)** host the majority of SpaceX launches.
- The Folium map provides an interactive visualization of launch site locations.

Interactive Marker Cluster Analysis



- Folium was used to visualize individual Falcon 9 launch records at **CCAFS SLC-40**.
- **Green markers** represent successful launches, while **red markers** represent failed launches.
- Marker clustering improves map readability when multiple launches occur at the same site.
- The marker cluster visualization improves readability and allows quick analysis of launch outcomes at CCAFS SLC-40.

Distance of Launch Sites from Nearby Landmarks

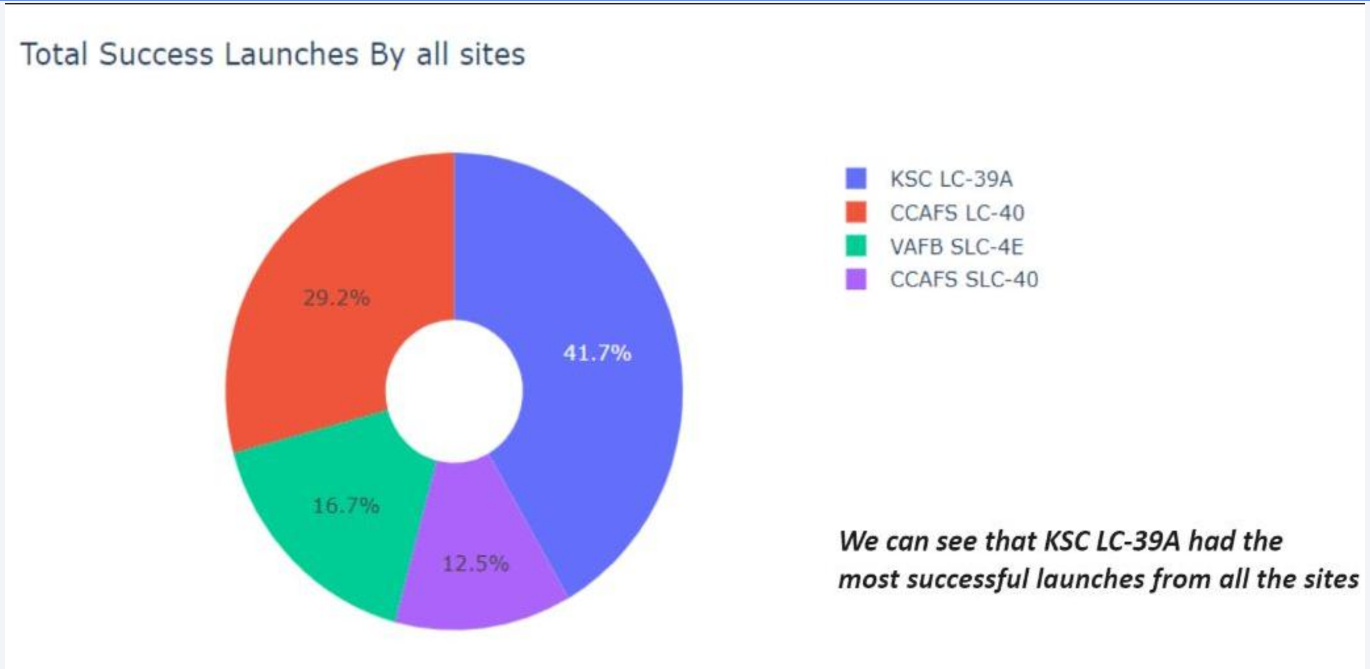


- The proximity analysis shows that SpaceX launch sites are strategically located near the coastline, enabling safer launch operations and efficient mission planning.

Section 4

Build a Dashboard with Plotly Dash

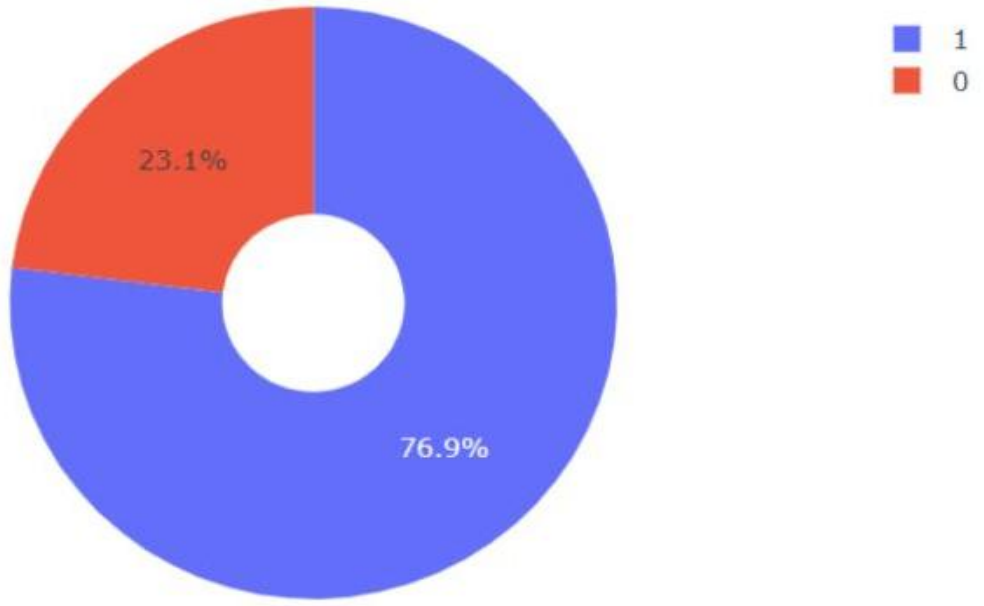
Interactive Dashboard (Plotly Dash)



- Interactive dashboard developed using Plotly Dash.
- Displays the distribution of successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.
- Interactive Plotly Dash dashboard illustrating the distribution of successful launches across all launch sites.

Highest Landing Success Rate by Launch Site

KSC LC-39A: Success vs Failure

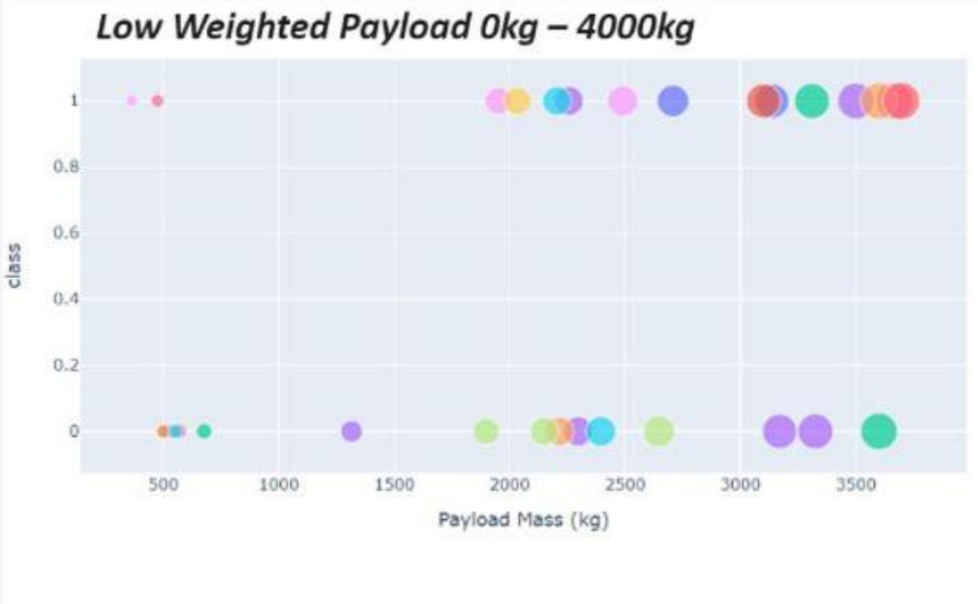


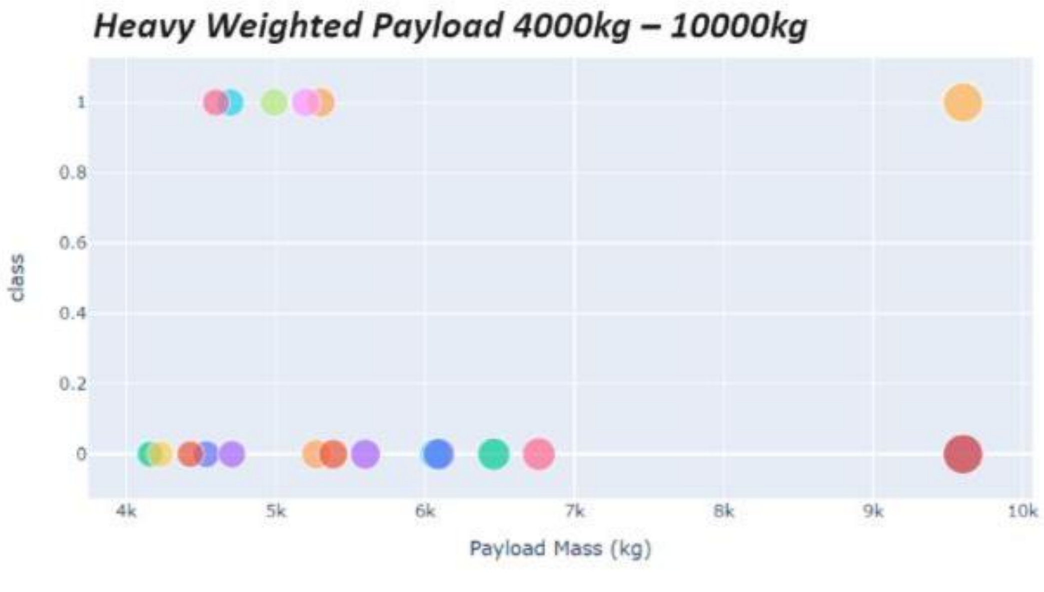
KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

- Shows the success and failure distribution for **KSC LC-39A**.
- Approximately **76.9%** of launches were successful.
- Helps compare mission outcomes at the selected launch site.
- KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

Payload Mass vs Launch Outcome

- The scatter plots show the relationship between payload mass and launch outcome.
- Most successful landings occurred for payloads between 2,000 and 4,000 kg.





Lower payloads generally achieved higher landing success rates than heavier payloads.

Section 5

Predictive Analysis (Classification)

Predictive Analysis (Classification)

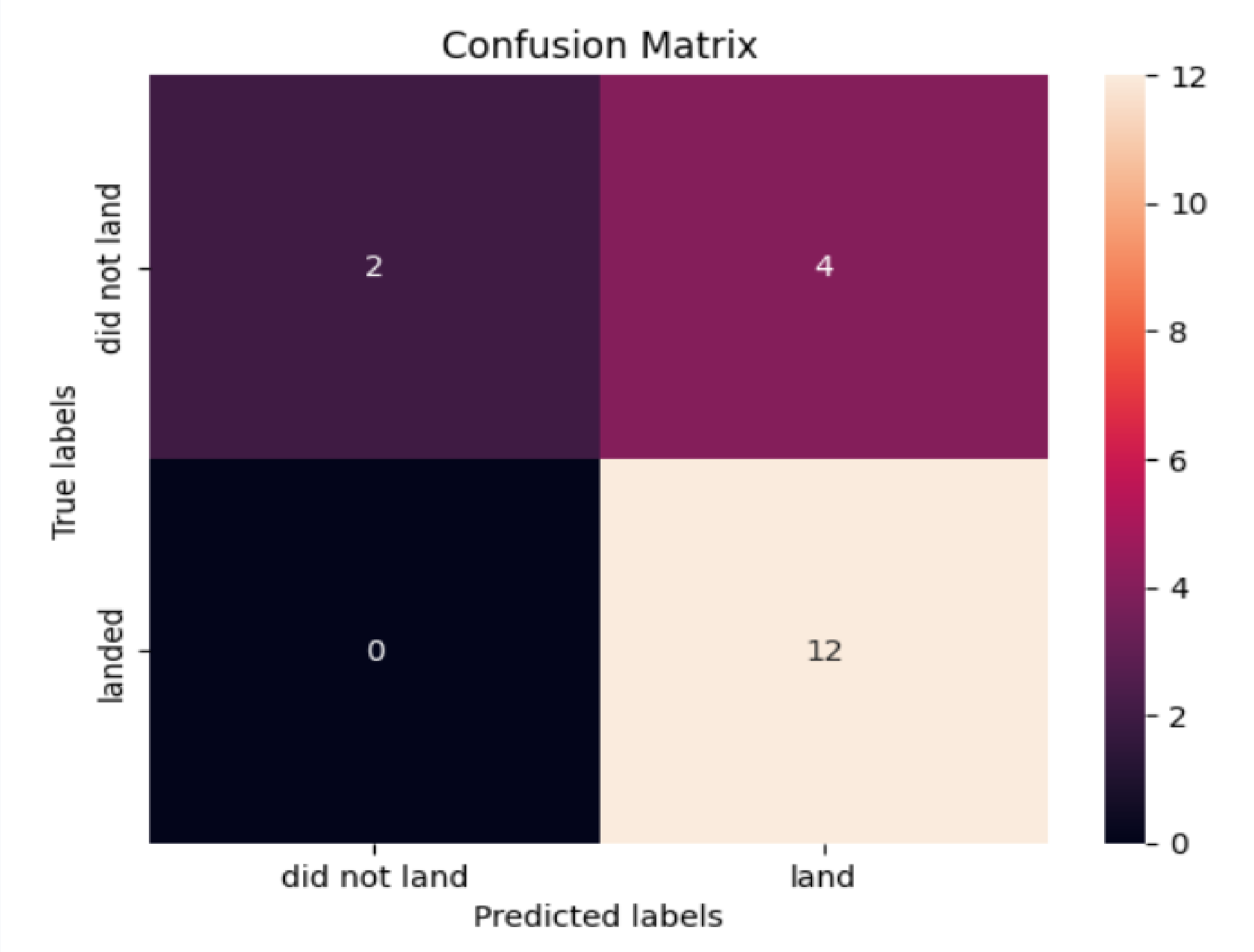
- Decision Tree achieved the highest classification accuracy (94.44%) among all evaluated machine learning models, making it the best-performing model for predicting Falcon 9 first-stage landing outcomes.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV

parameters = {
    'criterion': ('gini', 'entropy'),
    'max_depth': np.arange(2, 10),
    'max_features': ('auto', 'sqrt', 'log2')
}

tree = DecisionTreeClassifier()

tree_cv = GridSearchCV(
    estimator=tree,
    param_grid=parameters,
    cv=10
)

tree_cv.fit(X_train, Y_train)
```

```
print("Logistic Regression:", logreg_cv.score(X_test, Y_test))
print("SVM:", svm_cv.score(X_test, Y_test))
print("Decision Tree:", tree_cv.score(X_test, Y_test))
print("KNN:", knn_cv.score(X_test, Y_test))
```

```
Logistic Regression: 0.8333333333333334
SVM: 0.8333333333333334
Decision Tree: 0.9444444444444444
KNN: 0.7777777777777778
```

Confusion Matrix



- Compared multiple classification models for predicting Falcon 9 landing success.
- **Decision Tree** achieved the highest accuracy among the evaluated models.
- The selected model was used for predicting future landing outcomes.

Conclusions

- SpaceX has achieved a high success rate in Falcon 9 first-stage landings, demonstrating significant advancements in reusable rocket technology.
- Launch success was influenced by multiple factors, including launch site, payload mass, booster version, and landing method. Payloads between 2,000 and 4,000 kg generally showed higher landing success rates.
- Geospatial analysis showed that SpaceX launch sites are strategically located near coastal regions, supporting safer launch and recovery operations.
- Among the evaluated machine learning models, the Decision Tree classifier achieved the highest prediction accuracy (94.44%), making it the best-performing model for predicting Falcon 9 first-stage landing outcomes.
- Overall, this analysis demonstrates how data analytics, visualization, and machine learning can effectively support the prediction and evaluation of Falcon 9 first-stage landing success.

Thank you!